

Using Real Trees to Bring Nature Indoors

There is a growing movement worldwide to use plants in building design to establish closer connections with nature and soften hard city styles. The movement also seeks to improve designs to help protect the environment and increase personal wellness.

Many of the efforts by architects and designers have centered on one natural element: trees.

Examples of such design can be found in many places, including New York City. A building in Manhattan that houses the Ford Foundation has a 12-story light-filled entrance area, or atrium. It is also filled with plants and flowers including magnolias, eucalyptus, jacaranda, cryptomeria, iron bark and pear trees.

Another example sits within New York City's Brookfield Place shopping center. The center's main atrium includes a collection of 12-meter-tall Washingtonia palm trees.

In Singapore, the city-state's Jewel Changi airport offers travelers an inviting environment. It has 2,500 trees native to nations including Madagascar, Australia, Malaysia and Indonesia. The airport includes a 2.4-hectare indoor forest with walking paths.

An indoor waterfall is situated in a five-story high indoor forest filled with trees and plants at Changi Jewel Airport in Singapore. (Natalie Chin via AP)

Some apartments and mixed-use buildings are also putting trees and other greenery into their designs. Such buildings are sometimes called "greenscrapers," a term that comes from the word "skyscraper."

In Milan, Italy, architect Stefano Boeri designed Bosco Verticale, a 44-story building with 800 trees and other greenery. In the Netherlands, he also built the Trudo Vertical Forest. This is a low-income apartment building with trees growing on all sides.

People walk through the Tree Library park in Milan, Italy, Sunday, Dec. 9, 2018. (AP Photo/Luca Bruno)

Similar changes are also happening inside private homes. In some, trees are brought right indoors. In others, outdoor trees are carefully placed as important partners in the building plan.

Some architects even build homes around existing trees. Others plant one or more trees inside the home. And some architects or designers imagine natural settings indoors by using parts of trees. These designs are meant to create the look and feel of real wood, while bringing feelings of peace and calm.

A calm heart of the home

Ryoko Okada is an architect at New York City's Oda Architecture. She told The Associated Press that she and her co-workers aim to bring outdoor elements indoors as often as they can.

"There's nothing more calming than being surrounded by nature..." Okada said. She added that a living tree indoors can bring peace and happiness as people watch it grow and change.

Framing the views with existing trees

Another design example came from KAA Design Group in Los Angeles. A team from the company created a modern home that uses the property's existing trees, such as cork oaks, rare Torrey pines, and magnolias.

This image provided by KAA Design Group shows their Tree House project in California. The architect convince the owners to feature the old tree in the design of the house. (KAA Design Group via AP)

One very large, old cork oak sits in the center of the property. At first, the owners did not like the look and said they wanted the tree gone. One of the company's partners, Grant Kirkpatrick, had to persuade the homeowners to keep it. Kirkpatrick explained he succeeded by pointing out that the tree was "a 300-year-old antique that should be featured rather than removed."

The driveway on that property also offers a calming design. A black pine tree rises up from the driveway like a piece of natural art. In addition, views from almost every window in the house feature the beautiful nature outdoors.

Things to consider

Okada says she suggests people do not plant a full-size tree in their home without first seeking expert advice. She notes that a living tree will only do well if certain conditions – such as air, light and soil – are right.

Okada added that a living tree "...needs to be supported by proper infrastructure to stay alive and avoid home damage." She also urges people to heavily research a tree's natural environment to avoid drawing in unwanted pests.

Kirkpatrick noted that planting a tree indoors can be difficult. He explained that at least 1.2 meters of soil depth is needed, along with lots of natural light and the right temperature.

Other ways to use trees besides having a live one

Kirkpatrick said he worked on one property that experienced a serious fire in 2016. His team decided to use parts from one of the burned manzanita trees in the interior design. The piece was painted black and placed inside the property. He said it now represents an eye-catching, natural art piece.

I'm Anna Matteo.

Kim Cook reported this story for the Associated Press from New York, New York. Anna Matteo adapted it for VOA Learning English.

Vocabulary

skyscraper -n. ตึกระฟ้า, อาคารสูงระฟ้า

architect -n. สถาปนิก, ผู้ออกแบบอาคาร

apartment -n. ห้องชุด, แฟลต, อพาร์ทเมนต์

antique -adj. โบราณ, เก่าแก่, แบบสมัยก่อน

feature -n. ลักษณะเด่น, จุดเด่น, ส่วนที่สำคัญ

infrastructure -n. โครงสร้างพื้นฐาน, รากฐาน

pest -n. สัตว์รบกวน, ศัตรูพืช, แมลงศัตรู

Reading Comprehension Quiz

1. What is the primary goal of the growing movement to use plants in building design?

- A) To reduce construction costs.
- B) To promote modern architectural styles.
- C) To connect with nature, improve environment, and increase personal wellness.
- D) To only use plants that are native to the specific city.

2. What natural element has been the main focus of architects and designers in this movement?

- A) Flowers
- B) Water features
- C) Trees
- D) Herbs and shrubs

3. Which type of tree is NOT mentioned as being present in the Ford Foundation building's atrium in New York City?

- A) Magnolias
- B) Jacaranda
- C) Cherry
- D) Eucalyptus

4. What specific feature is highlighted in the main atrium of New York City's Brookfield Place shopping center?

- A) A vast indoor garden with various flowers.
- B) A collection of 12-meter-tall Washingtonia palm trees.
- C) An indoor waterfall.
- D) A simulated rainforest environment.

5. How large is the indoor forest at Singapore's Jewel Changi airport?

- A) 1,200 square meters
- B) 2.4 hectares
- C) 2,500 square meters
- D) 12 meters tall

6. What is the term used to describe apartment and mixed-use buildings that incorporate trees and other greenery into their designs?

- A) Ecoscrapers
- B) Biocrest
- C) Greenscrapers
- D) Naturescapes

7. Which architect designed both the Bosco Verticale in Milan and the Trudo Vertical Forest in the Netherlands?

- A) Ryoko Okada
- B) Grant Kirkpatrick
- C) Stefano Boeri
- D) Anna Matteo

8. According to the article, how are trees being incorporated into private homes?

- A) Only by planting them completely indoors.
- B) Only by carefully placing outdoor trees as partners in the building plan.
- C) By bringing trees indoors or placing outdoor trees as partners in the building plan.
- D) By replacing all indoor furniture with wooden elements.

9. What is one way architects and designers create a natural setting indoors, even without a live tree?

- A) By installing artificial trees.
- B) By painting walls green.

- C) By using parts of trees to create the look and feel of real wood.
- D) By playing nature sounds.

10. What does Ryoko Okada believe a living tree indoors can bring to people?

- A) Only aesthetic beauty.
- B) Peace and happiness as people watch it grow and change.
- C) Increased property value.
- D) A sense of obligation to care for it.

11. Why did the homeowners of the KAA Design Group project initially want to remove the large cork oak tree?

- A) It was diseased and dying.
- B) They disliked its appearance.
- C) It was blocking the sun.
- D) It was interfering with construction.

12. How did Grant Kirkpatrick convince the homeowners to keep the cork oak tree?

- A) He offered to pay for its relocation.
- B) He explained its environmental benefits.
- C) He highlighted its value as a '300-year-old antique.'
- D) He designed the house to hide it from view.

13. What is Ryoko Okada's main advice for people considering planting a full-size tree in their home?

- A) To ensure they have enough space.
- B) To seek expert advice first.
- C) To choose a fast-growing species.
- D) To only plant small, decorative trees.

14. Besides proper conditions, what else does Ryoko Okada suggest is crucial for a living indoor tree to thrive and avoid home damage?

- A) Daily watering and fertilization.
- B) Frequent pruning and shaping.
- C) Being supported by proper infrastructure.
- D) Protection from household pets.

15. How did Grant Kirkpatrick's team use parts of burned manzanita trees after a fire?

- A) They replanted them in a new location.

- B) They used them as structural support for the house.
- C) They turned them into eye-catching natural art pieces for the interior design.
- D) They converted them into firewood.